

Computer architecture and Organization

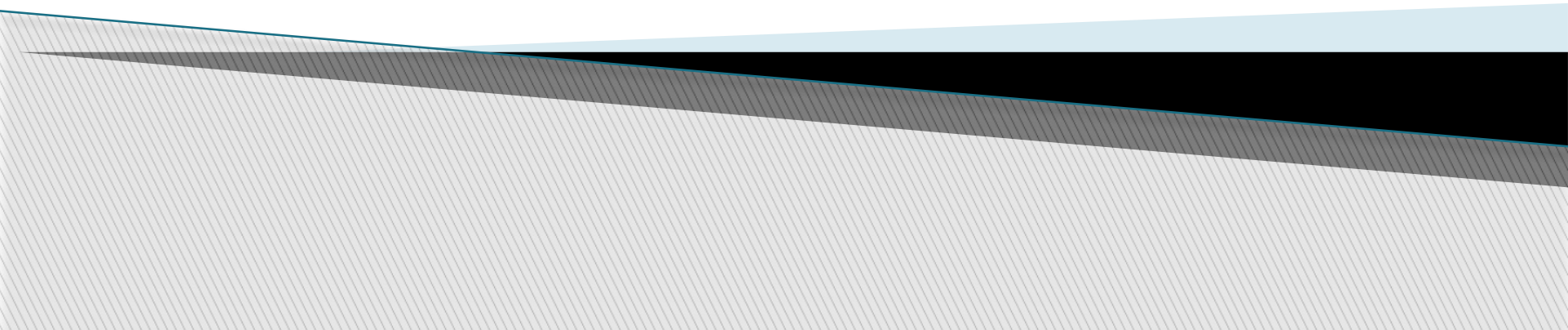
Composite Architectures

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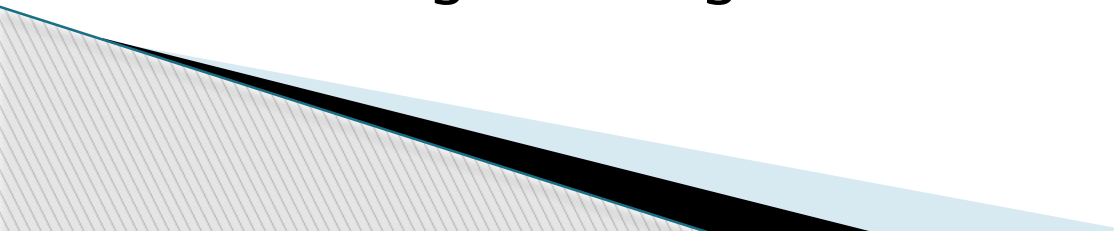
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The bottom of the slide features a decorative graphic consisting of several overlapping, wavy, horizontal bands. The colors transition from a light blue at the top, through a dark grey, to a light grey at the bottom. The waves are more pronounced on the left side and fade towards the right.

Composite Architecture

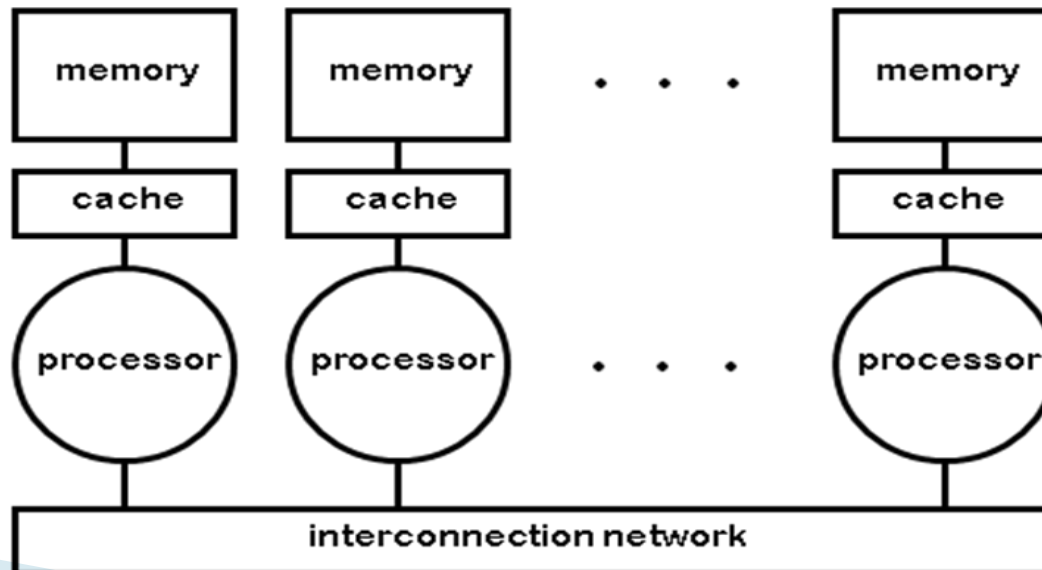
Multiprocessors:

- ▶ The Computer system having two or more processing units
 - ▶ Share main memory and peripherals
 - ▶ Reasons of using multiprocessors:
 - Multiple users.
 - Multiple applications.
 - Multitasking within an application
 - ▶ Two types of multiprocessor systems:
 - Message Passing, Shared memory
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Composite Architecture

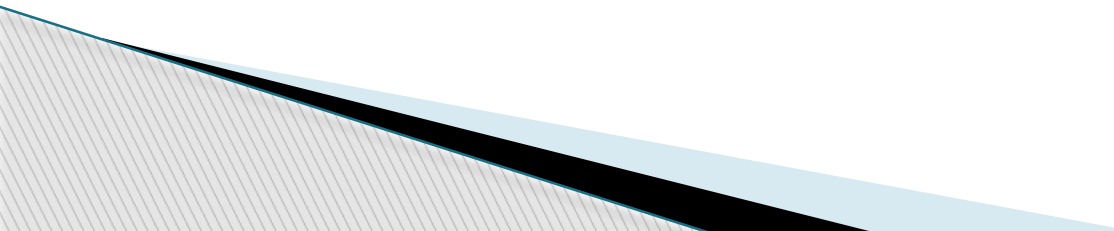
Message Passing

- Separate address space for each processor
- Processors communicate via message passing
- Processors have private memories



Composite Architecture

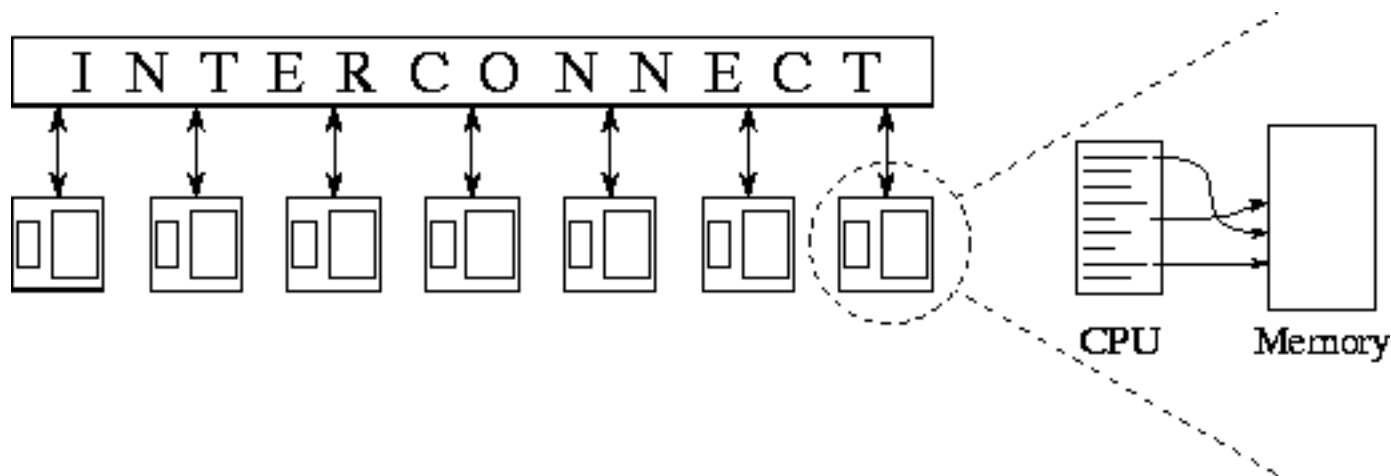
Multicomputers

- ▶ A computer made up of several computers.
 - ▶ Multicomputers are commonly used to get strong computing power.
 - ▶ A closely coupled processors that do not physically share memory
 - Cluster computers
 - Networks or clusters of computers
 - Can grow to a very large number of processors
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Composite Architecture

Multicomputer Consist of

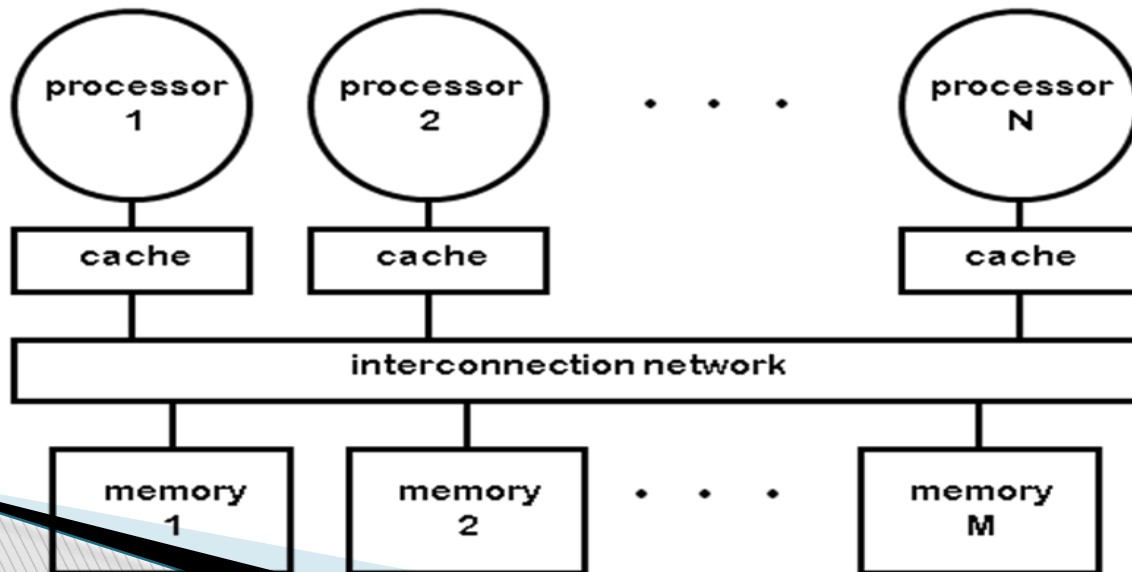
- Processing nodes – CPU, memory and network interface (NIC)
- I/O nodes – device controller and NIC
- Interconnection network



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Shared Memory

- Processors communicate with shared address space
- Processors communicate by memory read/write
- Easy on small-scale machines
- Lower latency



Composite Architecture

Comparison of CISC and RISC processors

CISC

Emphasis on hardware

Includes multi-clock complex instructions

Small code sizes, high cycles per second

Limited number of registers

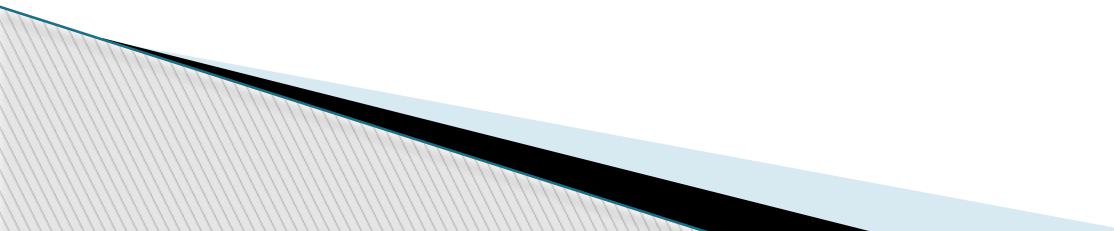
RISC

Emphasis on software

Single-clock, reduced instruction only

Low cycles per second, large code sizes

Large number of registers



Composite Architecture

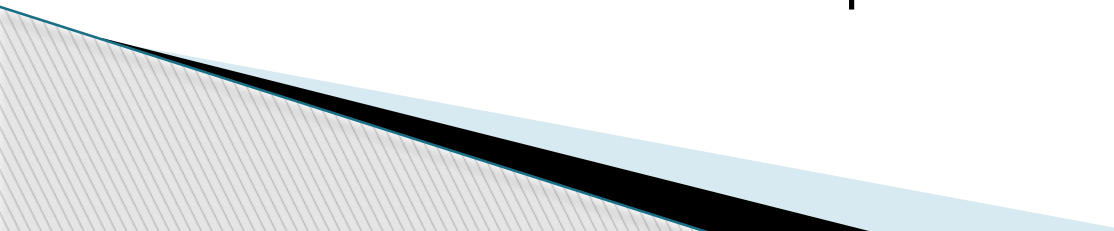
CISC –The PDP-11, 80x86 and Motorola 68K.

► Complex instruction set computer (CISC)

Several CISC characteristics:

- Complex instruction
- Efficient use of memory
- A small number of general purpose registers
- Several special purpose registers

Properties of CISC instructions:

- A 2-operand format, where instructions have a source and a destination.
 - Variable length instructions
 - Instructions which require multiple clock cycles to execute.
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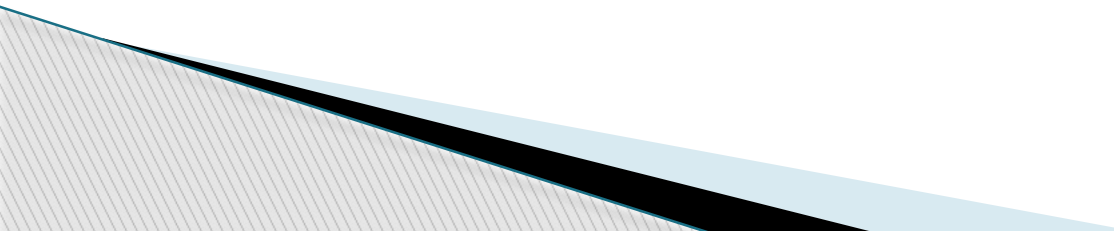
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RISC –IBM 801, Stanford MIPS, and Berkeley RISC
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- ▶ *Reduced Instruction Set Computer* (RISC) is a type of microprocessor architecture that utilizes
 - A small, highly-optimized set of instructions
 - A more specialized set of instructions
 - *One cycle execution time*: RISC processors have a CPI (clock per instruction) of one cycle
 - Supports *Pipelining*
 - *Large number of registers*: the RISC design philosophy generally incorporates a larger number of registers to prevent in large amounts of interactions with memory

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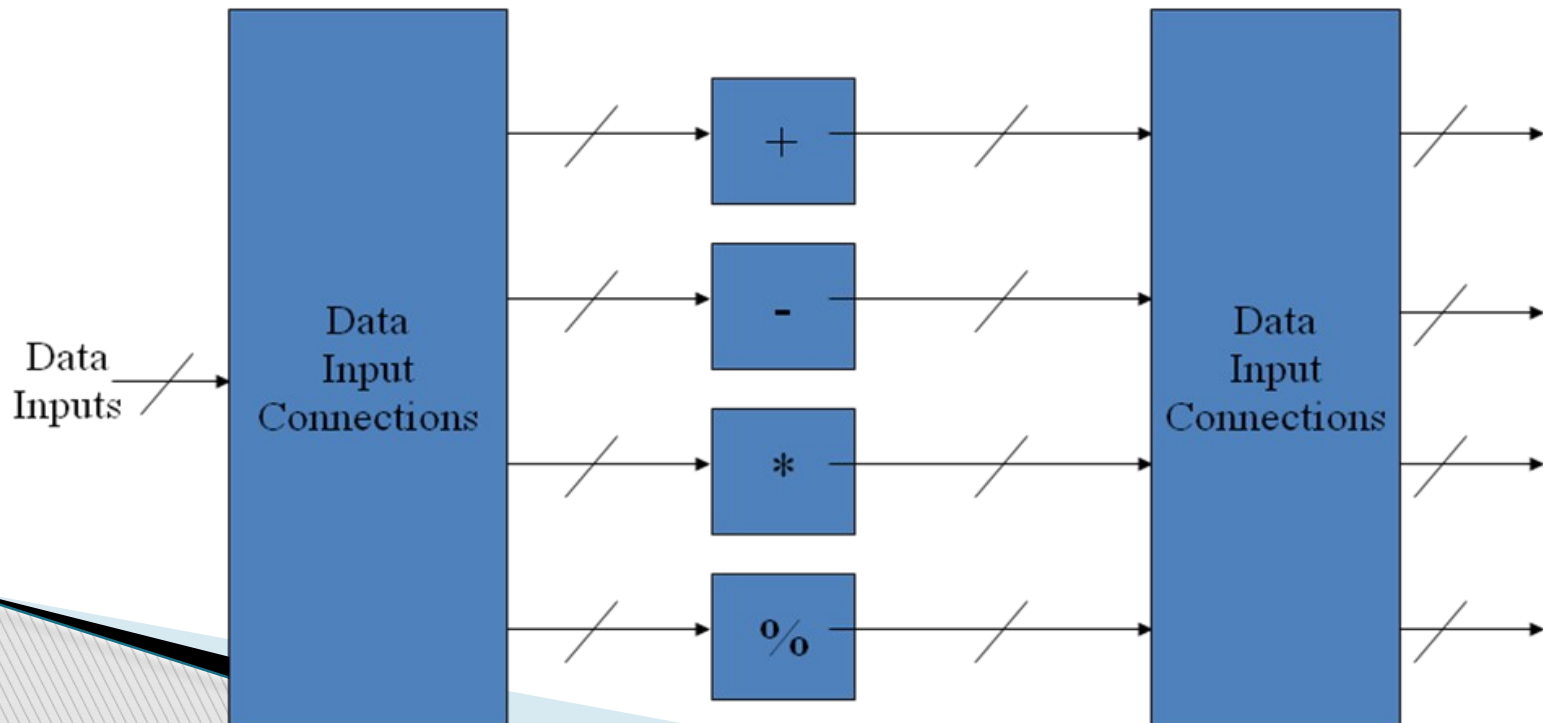
Parallel Processing

- ▶ Parallel processing – executing more than one instruction at the same time.
 - ▶ It is possible to achieve parallelism with a uniprocessor system.
 - ▶ The CPU must include a vectored arithmetic unit.
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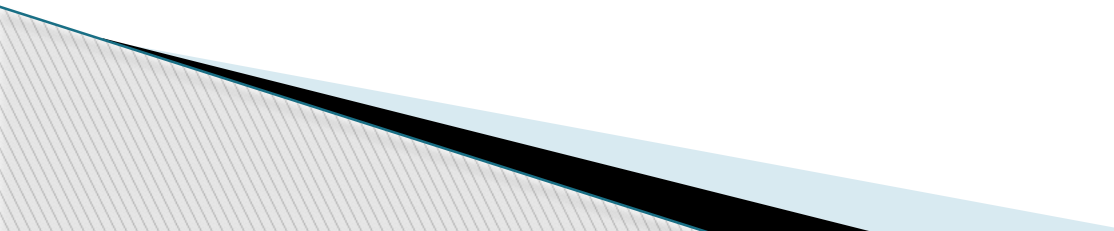
A vector arithmetic unit contains

- ▶ Multiple functional
- ▶ The control unit routes input values to the different functional units



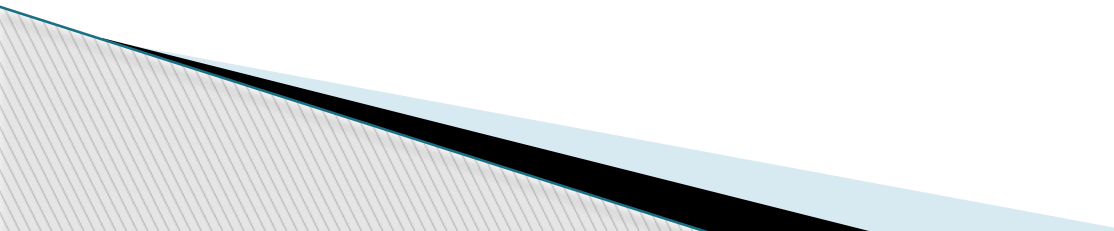
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Parallelism in multiprocessors

- ▶ Parallel processing systems achieve parallelism by having more than one processor performing tasks simultaneously.
 - ▶ Since multiprocessor systems are more complicated than uniprocessor systems, there are many different ways to organize the processors and memory.
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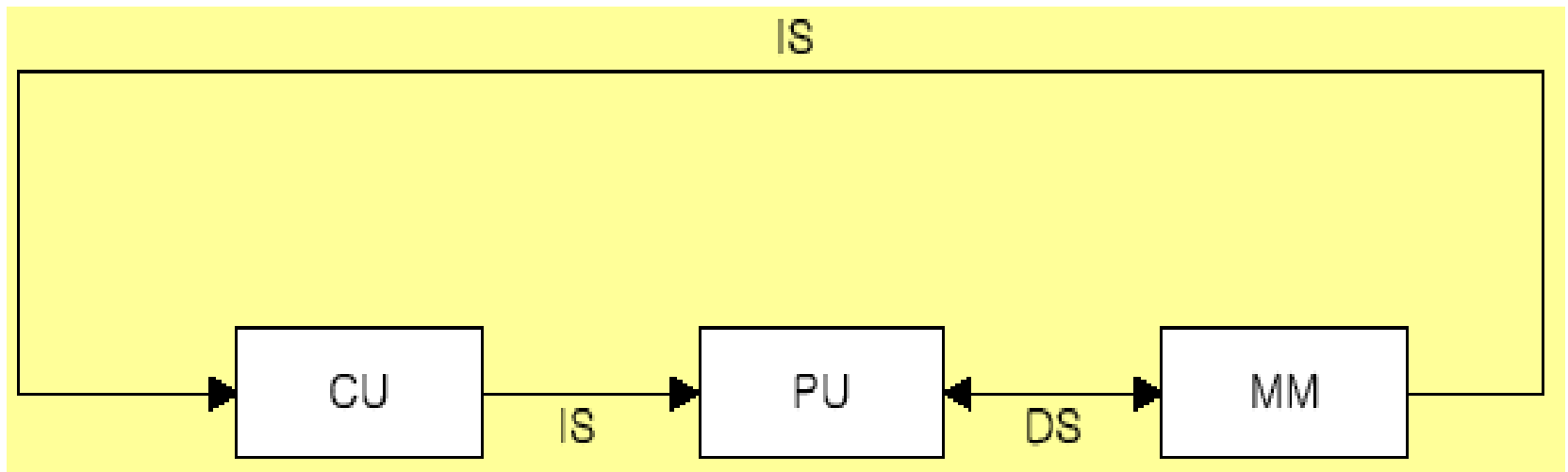
The Flynn's classification of parallel processing is as follows:

- ▶ **SISD:** Single instruction with single data – conventional computers
 - ▶ **SIMD:** Single instruction with multiple data – vector computing machines
 - ▶ **MISD:** Multiple instruction with single data – pipelining system
 - ▶ **MIMD:** Multiple instruction with multiple data – general purpose machine
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SISD

- ▶ Practical purpose: one processor is useful

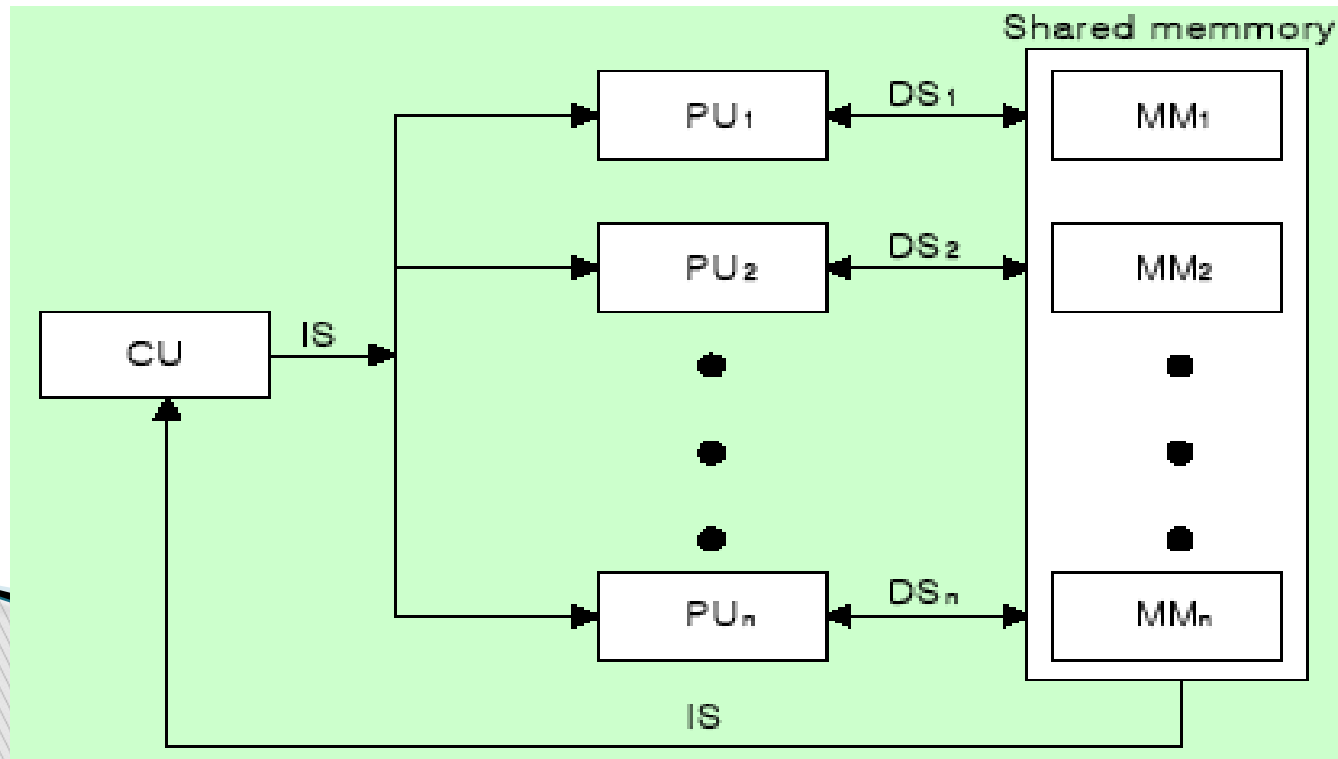


- ▶ IS → Instruction stream, DS → Data stream, CU → Control unit, PU → Processor unit, MM → Memory module

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SIMD

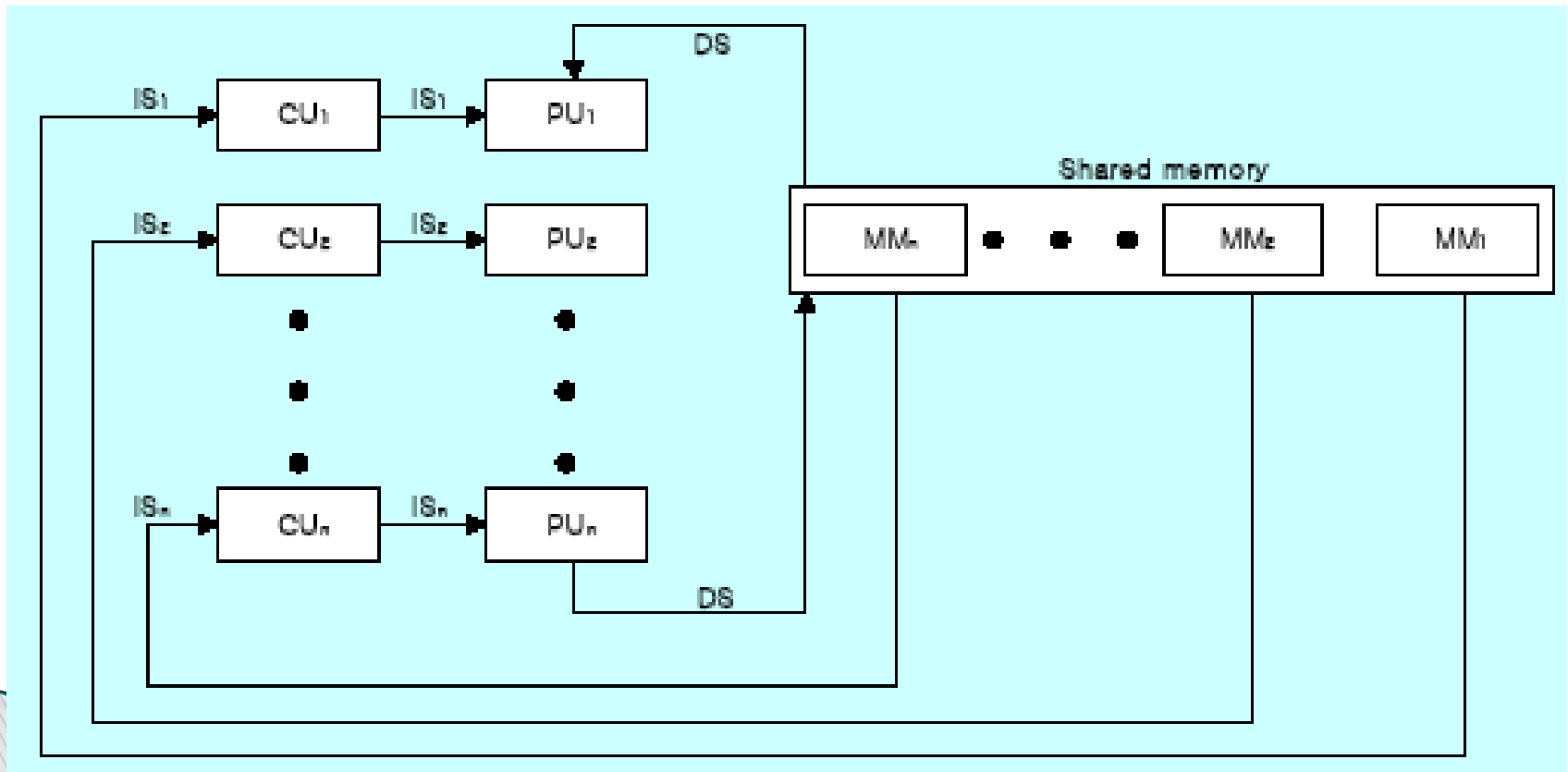
- ▶ Vector or array operations
 - one vector operation includes many operations on a data stream



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MISD

- ▶ Data Stream bottle neck



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MIMD

- Multiprocessor System
- Shared memory or Message passing

